Rahul Prasad

Senior Frontend Engineer · React.js · Next.js · TypeScript · Frontend System Design & Architecture

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EXECUTIVE SUMMARY

Senior Frontend Engineer with 7 years of experience architecting, scaling, and optimizing high-traffic web platforms, micro-frontends, and design systems using React.js, Next.js, TypeScript, JavaScript (ES6+), Redux Toolkit, and Node.js (BFF). Proven 0-to-1 and platform-scale ownership across enterprise compensation systems (120,000+ users), global digital platforms (1.5M+ monthly events across 15+ countries), and consumer banking applications (Citibank North America). Drives engineering velocity through runtime Module Federation, Core Web Vitals optimization (LCP, INP, CLS), Test-Driven Development (Jest, RTL, Cypress), Docker/Kubernetes CI/CD pipelines, and AI-augmented workflows (Claude, GitHub Copilot, MCP).

TECHNICAL SKILLS

- **Frontend Architecture & State Management:** React.js (Hooks, Custom Hooks, Context API, Suspense), Next.js, Redux, Redux Toolkit, Micro-Frontends (Webpack Module Federation), Single-Page Apps (SPA), Design Systems (60+ Components), Design Tokens (Style Dictionary), Angular, RxJS, Chart.js
- **Languages, Web Standards & Accessibility:** TypeScript, JavaScript (ES6+), HTML5, CSS3, SCSS, Browser APIs (Storage, History, Visibility & Event Telemetry), Responsive Design, WCAG 2.1 AA Accessibility (Lighthouse 90+, Axe, WAVE)
- **Backend-for-Frontend (BFF), APIs & Data Systems:** Node.js, Next.js API Routes (BFF), RESTful APIs, Microservice API Aggregation, Supabase (PostgreSQL), Complex Interactive Data Grids, Bulk File Upload & Multi-Step Approval Engines
- **Web Performance & Observability:** Core Web Vitals Optimization (LCP, INP, CLS), Route-Level Code Splitting, Lazy Loading, Bundle Size Reduction, Render-Cycle Optimization, High-Volume Event Instrumentation
- **Automated Testing, CI/CD & AI Workflows:** Test-Driven Development (TDD), Jest, React Testing Library, Cypress (E2E), Docker, Kubernetes, Jenkins CI, Webpack, Git, Bitbucket PR Checks, Static Analysis (ESLint/Prettier), AI-Assisted Development (Claude, GitHub Copilot, MCP)

PROFESSIONAL EXPERIENCE

Kaiser Permanente — Senior Frontend Engineer (Official Title: Senior IT Engineer, Web Applications)

Jul 2025 – Present · Pune, India · Enterprise Platform (120,000+ Users) · React.js, Next.js, TypeScript, Redux Toolkit, Node.js (BFF)

- **Micro-Frontend Architecture, Shared UI Library & Core Web Vitals:** Re-architected a legacy Angular/jQuery approvals platform (12,000+ monthly approvals, 4,500+ managers) into React.js, Next.js, and TypeScript micro-frontends via runtime Webpack Module Federation so 3 teams deploy independently; published a Confluence component library adopted by 2 sibling teams, cut bundle size 30%, achieved WCAG 2.1 AA (90+ Lighthouse), and improved LCP (3.8 s → 1.4 s), INP (290 ms → 75 ms), and CLS (0.18 → 0.02).
- **0-to-1 Product Architecture & Rule Engine (Wage Estimator):** Owned full SDLC solo from VP requirements and functional specs to release, designing a unified React.js, TypeScript, and Redux Toolkit rule engine across 8 unions, 140+ job codes, and 8 regional markets with a role-restricted superuser mirror view for live scenario simulation; used by 30,000+ of 70,000+ eligible employees and cut support tickets 67%.
- **Backend-for-Frontend (BFF) API Aggregation & Behavioral Testing:** Co-designed a Next.js API Routes (Node.js) BFF layer aggregating and normalizing 4–6 backend microservice calls per page into a single payload, cutting average API latency from 480 ms to 310 ms; authored automated unit, integration, and E2E tests asserting real user workflows.
- **No-Code Policy Configurator & High-Volume Data Grids:** Engineered a backward-compatible drag-and-drop JSON schema configurator with multi-attribute filtering, large interactive data grids, bulk file uploads, and multi-step approval workflows used by 25+ authors to manage 180+ nested policies across 400+ pages; cut setup time from 1 week to 30 minutes with zero post-launch defects.

- **Centralized Personalization Engine & Kubernetes CI/CD Automation:** Co-engineered a 20+ attribute personalization engine and role-restricted impersonation system (cutting visibility complaints 78% from ~45 to <10/month and retiring 35+ QA accounts) while mentoring junior engineers; built a fault-tolerant Docker/Kubernetes CI/CD pipeline scaling releases from 1 manual deploy/week to 12+ automated deployments (3 hours → 40 minutes).

Publicis Sapient — Senior Associate, Frontend Engineering (Associate Technology L1)

Oct 2021 – Jul 2025 · Gurugram, India · Global Digital Platform (14 Languages, 15+ Countries, 120+ Forms) · React.js, TypeScript, Webpack, Jest, Cypress

- **Test-Driven Development (TDD) & CI Quality Gatekeeping:** Championed TDD and automated testing across the engineering team using Jest, React Testing Library, and Cypress enforced as mandatory merge gates in Jenkins CI and Bitbucket PR checks, increasing test coverage from 45% to 75% and reducing production bug leakage 30%.
- **SPA Architecture Migration & 60-Component Design System:** Engineered modular React.js and TypeScript components within a 60-component design system to transition the platform into a Single-Page Application (SPA); implemented route-level code splitting and lazy loading to shrink the JS bundle 32% and cut page load from 4.1 s to 3.2 s while mentoring 4 junior engineers.
- **Design-Token Build Compiler (14 Languages):** Architected a centralized design-token pipeline using a customized Style Dictionary and Webpack build script emitting dedicated per-language CSS bundles across 14 languages instead of a monolithic file, shrinking per-language CSS payload 40% and establishing the org-wide standard.
- **Browser State Persistence & 5x Funnel Conversion Lift:** Built consent-based frontend state capture and REST API persistence across 120+ multi-step inquiry forms to trigger pre-filled abandonment recovery workflows before tab closure, lifting form completion conversion fivefold from 5% to 25% over a 90-day pre/post study.
- **High-Volume Behavioral Telemetry & WCAG 2.1 AA Compliance:** Instrumented 1.5M+ monthly user interaction events across 15+ country sites for personalized recommendations (+22% CTR, +35% session duration); remediated 240+ Axe/WAVE violations across 80+ templates and authored a shared accessibility library, lifting WCAG 2.1 AA compliance from 62% to 95% and cutting screen-reader defects 60%.

Tata Consultancy Services — Software Engineer, Frontend (System Engineer)

Jul 2019 – Oct 2021 · Gurugram, India · Consumer Banking & Telemetry Products: Citibank North America & CNOPS · React.js, Next.js, REST APIs, Chart.js

- **Consumer FinTech Architecture (Citi Virtual Account Numbers):** Architected 14 modular, DRY React.js and Next.js components for Citibank North America's credit card virtual account number flow, decoupling the REST API data/state layer from views and writing automated unit tests to cut customer transaction processing time 30%.
- **Real-Time Network Telemetry Dashboards (MWC 2020):** Built reusable React.js and Chart.js data-visualization components turning high-volume network site logs into interactive table and chart alarm views for CNOPS, eliminating redundant API calls and delivering the live operator showcase for Mobile World Congress 2020.

SELECTED PRODUCT ENGINEERING PROJECTS

- **Ehsaan Khata (ehsaankhata.com):** Mobile-first peer-to-peer ledger web app where groups record and approve favors with real-time balance sync across 55+ active groups. Stack: Next.js, React.js, TypeScript, Supabase (PostgreSQL), Google OAuth, Vercel.
- **Sensei Mods (senseimods.com):** Direct-to-consumer e-commerce platform featuring a live, interactive custom PC-cable configurator and Razorpay checkout across 150+ custom orders. Stack: Angular 16, TypeScript, RxJS.

EDUCATION

Bachelor of Technology (B.Tech) in Computer Science and Engineering · Guru Gobind Singh Indraprastha University, New Delhi, India · *2015 – 2019*